SZABIST
University

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Institute of Science and Technology

ROBOTICS & ARTIFICIAL INTELLIGENCE DEPARTMENT

Total Marks:_____

Obtained Marks: _____

Computing Vision

Lab Task # 01

Submitted To: **Mr. Muhammad Azhar**

Student Name: **Faiza Khan**

Reg. Number: **23108109**

1) List at least 10 features / functional requirements of your project

Prepare a scenario/case study for your own project idea and provide feature list relevant to the proposed scenario.

ADD KEYWORDS FOR SPECIFIC SUBJECT AND PROJECT

2) Provide a feasibility study

Prepare a table mapping each requirement to available Tools/resources/APIs/packages/libraries that would help you implement.

Sr.	Requirements	Resources

3) Technical developer guide

Choose any three features from your feature list and provide a detailed, step-by-step technical guide on how to implement them.

E.g. Step by step procedure to install a required library, to configure an API, and to implement a requirement by using specific Tools/resources/APIs/packages/libraries.

4) Implementation

Using any ONE of the available tools/technology stack, implement the three selected features inside a small working project.

Submission Requirements:

- Push your completed mini project to your own GitHub repository.
- Include a short README file describing the added features and navigation flow.
- Prepare screenshots of your project code from android studio, output from emulator or separate device, and your git repository along with feasibility study to include in your submission document.

Solution:

Real-Time Hand Gesture Counter System

Scenario

A classroom lab wants a simple system to count raised hands during interactive sessions using webcam input.

Keywords

Computer Vision, Hand Detection, Contours, Convex Hull, Thresholding, Real-Time Processing, Image Segmentation.

Functional Requirements

1. Capture webcam feed
2. Convert to HSV
3. Skin color detection
4. Threshold segmentation
5. Contour detection
6. Convex hull detection
7. Count raised hand shapes
8. Display count
9. Real-time processing
10. Exit with ESC

Feasibility

Sr	Requirement	Tools
1	HSV conversion	cvtColor()
2	Skin detection	inRange()
3	Contour detection	findContours()
4	Convex hull	convexHull()
5	Display count	putText()

Technical Feasibility

Uses OpenCV built-in image segmentation tools.

Economic Feasibility

Free and open-source.

Operational Feasibility

Simple webcam-based interaction.

Technical Guide

Feature 1: HSV Conversion

hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)

Feature 2: Skin Mask

mask = cv2.inRange(hsv, lower_skin, upper_skin)

Feature 3: Contour Counting

contours,_ = cv2.findContours(mask,cv2.RETR_TREE,cv2.CHAIN_APPROX_SIMPLE)

```
import cv2
import numpy as np

cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read()
    hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)

    lower_skin = np.array([0,20,70],dtype=np.uint8)
    upper_skin = np.array([20,255,255],dtype=np.uint8)

    mask = cv2.inRange(hsv,lower_skin,upper_skin)

    contours,_ = cv2.findContours(mask,
                                  cv2.RETR_TREE,
                                  cv2.CHAIN_APPROX_SIMPLE)

    count = len(contours)
```

```
    cv2.putText(frame,"Hands: "+str(count),
                (10,50),
                cv2.FONT_HERSHEY_SIMPLEX,
                1,(0,255,0),2)

    cv2.imshow("Hand Counter",frame)

    if cv2.waitKey(1)==27:
        break

cap.release()
cv2.destroyAllWindows()
```

HandGestureCounter Public

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23108109-cmd Add files via upload 87ec45a · now 3 Commits

README.md	Update README.md	1 minute ago
app.py	Add files via upload	now

README

Hand Gesture Counter System

Description

Counts raised hands using skin color detection and contour detection.

Features

- HSV conversion
- Skin segmentation
- Contour detection
- Real-time counter display
- Webcam-based interaction

Technologies

- Python
- OpenCV
- NumPy